Lecture 1

Continued fractions and matrix fixed points

One repeating block, one quadratic equation

$$x = \frac{ax + b}{cx + d}$$

A repeating pattern

$$\sqrt{2} = [1; 2, 2, 2, \ldots] = [1; \overline{2}]$$
$$1 + \sqrt{2} = [2; 2, 2, 2, \ldots] = [\overline{2}]$$

The first expansion is eventually periodic.

The second is purely periodic.

The two periodicity theorems

A quadratic irrational is an irrational real root of an integer quadratic.

Lagrange

$$\text{eventually periodic} \iff \text{quadratic irrational}$$

Galois

A quadratic irrational x is purely periodic exactly when

$$x > 1 \qquad \text{and} \qquad -1 < x' < 0.$$

Here x' is the other root of its quadratic equation.

A Möbius transformation

$$M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \qquad T_M(t) = \frac{at+b}{ct+d}$$

$$a, b, c, d \in \mathbb{Q}, \qquad ad - bc \neq 0$$

We use the formula where $ct + d \neq 0$.

The determinant condition makes the transformation invertible.

Why a matrix acts on a number

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} t \\ 1 \end{pmatrix} = \begin{pmatrix} at+b \\ ct+d \end{pmatrix}$$

Take the ratio of the two output coordinates.

$$t \longmapsto \frac{at+b}{ct+d}$$

Multiplying every entry of M by the same nonzero number leaves T_M unchanged.

Composition

$$T_M \circ T_N = T_{MN}$$

The rightmost matrix acts first.

$$M = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad N = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$T_M(T_N(t)) = 1 + \frac{1}{t}, \qquad MN = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$$

An inverse, by solving for t

$$y = \frac{at+b}{ct+d} \qquad \Longrightarrow \qquad t = \frac{dy-b}{a-cy}$$

$$M^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

$$(T_M)^{-1} = T_{M^{-1}}$$

The scalar factor disappears when we take a coordinate ratio.

One continued-fraction digit

$$D(a) = \begin{pmatrix} a & 1 \\ 1 & 0 \end{pmatrix}$$

$$T_{D(a)}(t) = a + \frac{1}{t}$$

The matrix inserts one digit before the remaining tail.

$$\det D(a) = -1, \qquad D(a)^{-1} = \begin{pmatrix} 0 & 1 \\ 1 & -a \end{pmatrix}$$

A block of digits

$$[a_0; a_1, \ldots, a_{r-1}, t] = T_{D(a_0)D(a_1)\cdots D(a_{r-1})}(t)$$

For the block $1, 2$,

$$D(1)D(2) = \begin{pmatrix} 3 & 1 \\ 2 & 1 \end{pmatrix}$$

$$1 + \frac{1}{2 + 1/t} = \frac{3t + 1}{2t + 1}.$$

A repeating tail

$$x = [\overline{b_0, \ldots, b_{\ell-1}}], \qquad b_i \geq 1$$

$$N = D(b_0) \cdots D(b_{\ell-1}) = \begin{pmatrix} A & B \\ C & D_0 \end{pmatrix}$$

Removing one whole period leaves the same tail.

$$x = T_N(x) = \frac{Ax + B}{Cx + D_0}$$

Here $B, C > 0$ and $\det N = (-1)^\ell$.

A fixed point gives an equation

$$x = \frac{Ax + B}{Cx + D_0}$$

$$Cx^2 + D_0 x = Ax + B$$

$$Cx^2 + (D_0 - A)x - B = 0$$

Since $C > 0$, this is a genuine quadratic equation.

Thus a purely periodic simple continued fraction is quadratic.

Your turn: the block $1, 2$

$$x = [\overline{1, 2}], \qquad N = \begin{pmatrix} 3 & 1 \\ 2 & 1 \end{pmatrix}$$

Find a quadratic equation for x.

Find both roots.

Which root can be the continued fraction?

Take one minute before we compare answers.

The periodic root

$$2x^2 - 2x - 1 = 0$$

$$x = [\overline{1, 2}] = \frac{1 + \sqrt{3}}{2}$$

The other root is

$$x' = \frac{1 - \sqrt{3}}{2}, \qquad -1 < x' < 0.$$

A finite prefix

$$x = T_N(x), \qquad y = T_M(x)$$

The original number y may have extra digits before the period.

$$y = [1; \overline{1, 2}] = 1 + \frac{1}{x}, \qquad M = D(1).$$

Which transformation fixes y?

The prefix conjugation identity

$$\begin{aligned} y &= T_M(x) \\ &= T_M\big(T_N(x)\big) \\ &= T_M\big(T_N(T_{M^{-1}}(y))\big). \end{aligned}$$

$$y = T_{MNM^{-1}}(y)$$

Attaching a prefix conjugates the transformation that fixes the tail.

Why the new equation is genuinely quadratic

Put $K = MNM^{-1}$.

K has integer entries and determinant ± 1.

K is nonscalar, since conjugating a scalar matrix gives a scalar matrix.

y is irrational, since $T_{M^{-1}}(y) = x$.

For $K = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$,

$$cy^2 + (d - a)y - b = 0.$$

This polynomial is nonzero and cannot be merely linear.

The equation for $\sqrt{3}$

$$M = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}, \qquad N = \begin{pmatrix} 3 & 1 \\ 2 & 1 \end{pmatrix}, \qquad M^{-1} = \begin{pmatrix} 0 & 1 \\ 1 & -1 \end{pmatrix}$$

$$MNM^{-1} = \begin{pmatrix} 2 & 3 \\ 1 & 2 \end{pmatrix}$$

$$y = \frac{2y+3}{y+2} \qquad \Longrightarrow \qquad y^2 = 3$$

$$[1; \overline{1, 2}] = \sqrt{3}$$

Your turn: a different prefix

Keep $x = [\overline{1, 2}]$, but use $M = D(2)$.

$$y = [2; \overline{1, 2}] = 2 + \frac{1}{x}$$

Compute MNM^{-1} and identify y.

Can you write its digits with a bar starting at the first digit?

A written prefix need not be a genuine preperiod.

A prefix that disappears

$$D(2)ND(2)^{-1} = \begin{pmatrix} 3 & 2 \\ 1 & 1 \end{pmatrix}$$

$$y^2 - 2y - 2 = 0, \qquad y = 1 + \sqrt{3}$$

$$[2; \overline{1, 2}] = [\overline{2, 1}]$$

The digits repeat from the beginning after all.

Quadratic conjugates

$$x = u + v\sqrt{D}, \qquad x' = u - v\sqrt{D} \qquad (u, v \in \mathbb{Q},\ v \neq 0)$$

Replacing $\sqrt{D}$ by $-\sqrt{D}$ respects rational arithmetic.

$$\big(T_M(x)\big)' = T_M(x')$$

For example,

$$(\sqrt{2})' = -\sqrt{2}, \qquad (1 + \sqrt{2})' = 1 - \sqrt{2}.$$

Reduced surds

A quadratic irrational is reduced when

$$x > 1 \qquad \text{and} \qquad -1 < x' < 0.$$

x	x'	Reduced?
$\sqrt{2}$	$-\sqrt{2}$	No
$1+\sqrt{2}$	$1-\sqrt{2}$	Yes
$\sqrt{2}/2$	$-\sqrt{2}/2$	No

What we have proved

$$\text{eventually periodic} \quad \Longrightarrow \quad \text{quadratic irrational}$$

A periodic tail satisfies a fixed-point equation.

A finite prefix changes N to MNM^{-1}.

Next lecture: why every quadratic irrational repeats, and why reduced surds repeat immediately.

Exit question: Why does $t = T_I(t)$ prove nothing?

Composition in coordinates

$$M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \qquad N = \begin{pmatrix} e & f \\ g & h \end{pmatrix}$$

$$\begin{aligned} T_M(T_N(t)) &= \frac{a(et+f)+b(gt+h)}{c(et+f)+d(gt+h)} \\ &= \frac{(ae+bg)t+(af+bh)}{(ce+dg)t+(cf+dh)}. \end{aligned}$$

These are exactly the four entries of MN.

The nonscalar fixed-point lemma

Let $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ be an invertible rational matrix.

$$t = T_M(t) \quad \Longrightarrow \quad ct^2 + (d-a)t - b = 0.$$

If all coefficients vanish, then $M = aI$.

If $c = 0$ and the polynomial is nonzero, its root is rational.

Therefore an irrational fixed point of a nonscalar rational matrix is a quadratic irrational.

Why an infinite simple continued fraction is irrational

The convergents are in lowest terms, $q_k \to \infty$, and

$$\left| x - \frac{p_k}{q_k} \right| \leq \frac{1}{q_k q_{k+1}}.$$

If $x = p/q$ were rational, then eventually $p_k/q_k \neq p/q$, and

$$\left| \frac{p}{q} - \frac{p_k}{q_k} \right| \geq \frac{1}{q q_k}.$$

For $q_{k+1} > q$, these inequalities contradict each other.